Simulated Power Plant Reactor System

System and Software Requirements Specification

Research and Training Simulation
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Abstract

This specification defines functional, operational, interface, data, performance, security, and verification requirements for a fictional reactor-based power-plant simulator. It is intended for requirements engineering, software safety analysis, human–AI evaluation, and classroom experiments. The document deliberately presents requirements in a uniform format without assigning safety classifications. The modeled plant is a simplified educational abstraction; this document must not be used to design, license, operate, or modify an actual nuclear facility.

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1 Introduction

1.1 Purpose

The purpose of this document is to specify the externally observable behavior and supporting software capabilities of a simulated power-plant reactor system. Requirements are written as individual, numbered statements to support extraction, traceability, review, classification, and testing.

1.2 Scope

The system is a software-only simulator of a fictional pressurized-water-reactor-style plant, including reactor physics abstractions, thermal-fluid processes, electrical generation, operator controls, protection logic, training scenarios, logging, and post-run analysis. No physical plant equipment is controlled.

1.3 Intended Audience

Software engineers, requirements engineers, simulation developers, instructors, human-factors researchers, verification engineers, and participants in controlled safety-analysis studies.

1.4 Document Conventions

Each normative statement uses `REQ-AAA-NNN`, where `AAA` is a subsystem code and `NNN` is a three-digit sequence number beginning at 001 within that subsystem. Each requirement is expressed with "shall." Requirement identifiers are stable within this document version. Requirements are not annotated with any safety classification or answer key.

1.5 Research Use and Boundaries

The specification contains design choices and operational behaviors for a fictional training system. Its statements are not claims of regulatory compliance, industry best practice, or engineering suitability for a real reactor. Some requirements may need reconsideration in a later engineering review; the purpose of the present document is to supply unclassified source material for independent analysis.

2 System Overview

2.1 Concept of Operations

An instructor configures a scenario and initial plant state. A trainee starts a run, observes simulated process variables, issues permitted commands, responds to events, and ends the session. The simulator records state histories, operator actions, and subsystem events for replay and evaluation.

2.2 Operating Modes

The supported modes are initialization, startup, normal operation, planned shutdown, maintenance exercise, and emergency exercise. Mode-specific logic is defined by the simulator configuration rather than by any physical equipment.

2.3 System Context and Interfaces

The simulation engine exchanges state variables with subsystem models; the operator interface sends commands to the engine and receives measurements, alarms, and status. Scenario management injects

disturbances. Logging and replay persist run artifacts. User management provides identity and role information.

2.4 Modeling Assumptions

The reactor, coolant, steam, and electrical subsystems are simplified, configurable numerical models. Their outputs are illustrative rather than validated physical predictions. Thresholds and controller parameters are training-scenario settings; no real-world operating setpoints are prescribed here.

3 Requirement Identifier Register

Code	Subsystem	Count
SYS	Plant-Wide Simulation and Operating Modes	16
KIN	Reactor Core and Neutronics Model	14
CRD	Control Rod Drive Simulation	14
RCS	Reactor Coolant System	16
PRS	Pressurizer and Pressure Regulation	12
SGS	Steam Generator and Secondary Heat Transfer	14
TUR	Turbine and Generator	14
ELE	Electrical Power and Distribution	14
RPS	Reactor Protection System	16
ECC	Emergency Cooling and Mitigation	14
CNT	Containment and Environmental Monitoring	12
INS	Instrumentation and Sensor Data	16
ALM	Alarms, Events, and Notifications	16
HMI	Human-Machine Interface and Operator Commands	16
AUT	Automatic Control and Setpoint Management	14
SCN	Scenario Authoring and Fault Injection	16
DAT	Data Storage, Logging, and Replay	16
SEC	User Accounts and Access Control	14
NET	Software Interfaces and Communications	14
VNV	Verification, Validation, and Experiment Support	16
PER	Performance, Reliability, and Recovery	14
DOC	Documentation, Training, and Usability	12

4 Detailed System and Software Requirements

4.1 Plant-Wide Simulation and Operating Modes (SYS)

`REQ-SYS-001`
The simulator shall represent reactor, primary coolant, secondary heat transfer, electrical, instrumentation, and operator-interface subsystems as coupled software models.

`REQ-SYS-002`
The simulator shall support startup, normal operation, planned shutdown, maintenance, and emergency exercise modes.

`REQ-SYS-003`
The simulator shall assign a unique identifier and simulation timestamp to each run.

`REQ-SYS-004`
The simulator shall initialize every modeled subsystem to a documented state before advancing simulated time.

`REQ-SYS-005`
The simulator shall allow an instructor to select a predefined initial condition before a training session.

`REQ-SYS-006`
The simulator shall synchronize subsystem calculations at a configurable simulation interval.

`REQ-SYS-007`
The simulator shall display the currently selected operating mode on the operator dashboard.

`REQ-SYS-008`
The simulator shall allow the operator to change the plant operating mode from the main dashboard without an additional confirmation step.

`REQ-SYS-009`
The simulator shall record every mode transition with the initiating actor and preceding mode.

`REQ-SYS-010`
The simulator shall reject initial conditions that contain undefined numerical values.

`REQ-SYS-011`
The simulator shall expose a read-only summary of major plant state variables through an internal API.

`REQ-SYS-012`
The simulator shall permit an instructor to pause and resume simulated time.

`REQ-SYS-013`
The simulator shall preserve the last computed plant state while the simulation is paused.

`REQ-SYS-014`
The simulator shall allow a saved scenario to override subsystem initialization defaults.

`REQ-SYS-015`
The simulator shall provide a visible indication when any subsystem model is unavailable.

`REQ-SYS-016`
The simulator shall continue the current scenario when a noncritical visualization widget fails.

4.2 Reactor Core and Neutronics Model (KIN)

`REQ-KIN-001`
The core model shall compute a normalized reactor power estimate for each simulation step.

`REQ-KIN-002`
The core model shall represent changes in reactivity arising from modeled control inputs and feedback variables.

`REQ-KIN-003`
The core model shall maintain a history of calculated power values for the current run.

`REQ-KIN-004`
The core model shall display normalized power as a percentage of the scenario reference value.

REQ-KIN-005
The core model shall reject negative values for absolute neutron population state variables.

REQ-KIN-006
The core model shall allow instructor-defined disturbances to alter the simulated reactivity input.

REQ-KIN-007
The core model shall apply a configurable smoothing filter to power values displayed on the operator screen.

REQ-KIN-008
The core model shall use the most recent available temperature estimate when a new temperature sample is delayed.

REQ-KIN-009
The core model shall generate an event when normalized power exceeds a scenario-defined monitoring threshold.

REQ-KIN-010
The core model shall continue computing power using the last valid feedback value when a sensor feed is marked stale.

REQ-KIN-011
The core model shall publish its calculated power value to the thermal model each simulation step.

REQ-KIN-012
The core model shall document numerical integration tolerances in its configuration metadata.

REQ-KIN-013
The core model shall store the input parameters used to reproduce a completed run.

REQ-KIN-014
The core model shall allow instructor scripts to modify core-model parameters during a paused scenario.

4.3 Control Rod Drive Simulation (CRD)

REQ-CRD-001
The control rod model shall represent the commanded and indicated position of each modeled rod group.

REQ-CRD-002
The control rod model shall display rod-group position using a consistent normalized scale.

REQ-CRD-003
The control rod model shall enforce configured travel limits during ordinary positioning commands.

REQ-CRD-004
The control rod model shall record each rod-motion command with its source and timestamp.

REQ-CRD-005
The control rod model shall allow an operator to issue incremental insertion and withdrawal commands.

REQ-CRD-006
The control rod model shall indicate when commanded and modeled positions disagree beyond a configured tolerance.

REQ-CRD-007
The control rod model shall accept a new withdrawal command while a prior withdrawal command is still being processed.

REQ-CRD-008
The control rod model shall stop a simulated movement when its travel limit is reached.

REQ-CRD-009
The control rod model shall support instructor injection of a stuck-rod condition.

REQ-CRD-010
The control rod model shall display the selected rod group before executing a command.

REQ-CRD-011
The control rod model shall preserve the previous commanded position when its command input is malformed.

REQ-CRD-012
The control rod model shall permit movement commands to remain queued during a temporary loss of operator connectivity.

REQ-CRD-013
The control rod model shall publish rod-position feedback to the core model at each simulation step.

REQ-CRD-014
The control rod model shall support a scripted sequence of rod movements for repeatable exercises.

4.4 Reactor Coolant System (RCS)

REQ-RCS-001
The coolant model shall calculate simulated primary-loop temperature, pressure, and mass-flow state variables.

REQ-RCS-002
The coolant model shall exchange heat-transfer estimates with the core and steam-generator models.

REQ-RCS-003
The coolant model shall present inlet and outlet temperature trends on the operator dashboard.

REQ-RCS-004
The coolant model shall detect a modeled reduction in coolant flow relative to a scenario-defined threshold.

REQ-RCS-005
The coolant model shall retain the most recent flow reading for display during a brief data-link interruption.

REQ-RCS-006
The coolant model shall model pump start, stop, and degraded-performance states.

REQ-RCS-007
The coolant model shall generate a timestamped event when a pump state changes.

REQ-RCS-008
The coolant model shall allow an operator to stop a simulated coolant pump from the equipment panel.

REQ-RCS-009
The coolant model shall display a warning when pressure data is unavailable for more than the configured freshness interval.

REQ-RCS-010
The coolant model shall accept a pump-start command even when the flow sensor is reporting an invalid

value.

REQ-RCS-011
The coolant model shall limit computed temperature values to the numeric range supported by the simulation engine.

REQ-RCS-012
The coolant model shall provide coolant inventory estimates to the emergency cooling model.

REQ-RCS-013
The coolant model shall support instructor-controlled leakage disturbances.

REQ-RCS-014
The coolant model shall save pressure and temperature histories for post-run analysis.

REQ-RCS-015
The coolant model shall show the active pump configuration in the scenario summary.

REQ-RCS-016
The coolant model shall apply configuration changes to the flow model at the next simulation step.

4.5 Pressurizer and Pressure Regulation (PRS)

REQ-PRS-001
The pressure regulation model shall compute a simulated primary-loop pressure estimate.

REQ-PRS-002
The pressure regulation model shall represent heater and spray actuation as distinct simulated control inputs.

REQ-PRS-003
The pressure regulation model shall display pressure setpoint and measured pressure together.

REQ-PRS-004
The pressure regulation model shall generate a deviation event when simulated pressure differs from its setpoint beyond a configured tolerance.

REQ-PRS-005
The pressure regulation model shall allow operators to change the pressure setpoint within scenario-defined bounds.

REQ-PRS-006
The pressure regulation model shall apply the most recent setpoint immediately after a mode change.

REQ-PRS-007
The pressure regulation model shall support instructor injection of a stuck-actuator condition.

REQ-PRS-008
The pressure regulation model shall preserve the last pressure estimate when an upstream model update is missed.

REQ-PRS-009
The pressure regulation model shall report actuator-command history to the event recorder.

REQ-PRS-010
The pressure regulation model shall permit automatic pressure regulation to remain active during a sensor-quality warning.

`REQ-PRS-011`
The pressure regulation model shall label pressure values with their configured engineering units.

`REQ-PRS-012`
The pressure regulation model shall provide a read-only diagnostic view of the active regulation state.

4.6 Steam Generator and Secondary Heat Transfer (SGS)

`REQ-SGS-001`
The steam-generator model shall estimate heat transfer between primary and secondary simulated loops.

`REQ-SGS-002`
The steam-generator model shall represent secondary-side water inventory as a time-varying state variable.

`REQ-SGS-003`
The steam-generator model shall display steam flow and feedwater flow as separate measurements.

`REQ-SGS-004`
The steam-generator model shall publish heat-removal estimates to the reactor thermal model.

`REQ-SGS-005`
The steam-generator model shall support a scenario in which feedwater delivery is reduced.

`REQ-SGS-006`
The steam-generator model shall record changes to feedwater control commands.

`REQ-SGS-007`
The steam-generator model shall allow the operator to switch between automatic and manual feedwater control.

`REQ-SGS-008`
The steam-generator model shall retain the last automatic controller output when changing to manual mode.

`REQ-SGS-009`
The steam-generator model shall indicate when secondary inventory falls outside a configured operating band.

`REQ-SGS-010`
The steam-generator model shall suppress duplicate low-inventory notifications within a configurable interval.

`REQ-SGS-011`
The steam-generator model shall represent the effect of a failed feedwater valve on simulated flow.

`REQ-SGS-012`
The steam-generator model shall include its model version in exported run metadata.

`REQ-SGS-013`
The steam-generator model shall support trend display over a user-selectable time window.

`REQ-SGS-014`
The steam-generator model shall accept scenario-specific controller parameters.

4.7 Turbine and Generator (TUR)

`REQ-TUR-001`
The turbine model shall calculate a simulated mechanical power value from secondary steam conditions.

`REQ-TUR-002`
The generator model shall calculate simulated electrical output from turbine power and generator status.

`REQ-TUR-003`
The turbine model shall represent turbine trip and restart states.

`REQ-TUR-004`
The turbine model shall display turbine speed and generator output on the operator dashboard.

`REQ-TUR-005`
The turbine model shall record operator-initiated turbine control actions.

`REQ-TUR-006`
The turbine model shall permit a turbine restart command before all advisory notifications have been acknowledged.

`REQ-TUR-007`
The generator model shall publish its power output to the electrical distribution model.

`REQ-TUR-008`
The turbine model shall support instructor injection of a load-rejection event.

`REQ-TUR-009`
The turbine model shall display the current turbine operating mode.

`REQ-TUR-010`
The turbine model shall limit numerical output to the configured model range.

`REQ-TUR-011`
The generator model shall maintain a history of output power values.

`REQ-TUR-012`
The turbine model shall allow automatic load-following commands when enabled by the scenario.

`REQ-TUR-013`
The turbine model shall use the most recent valid steam-flow value if a new sample is not available.

`REQ-TUR-014`
The generator model shall indicate when simulated synchronization conditions are not met.

4.8 Electrical Power and Distribution (ELE)

`REQ-ELE-001`
The electrical model shall represent normal supply, alternate supply, and loss-of-supply conditions.

`REQ-ELE-002`
The electrical model shall indicate the active simulated power source for each modeled bus.

`REQ-ELE-003`
The electrical model shall record source-transfer commands and resulting states.

`REQ-ELE-004`
The electrical model shall calculate a simplified balance between simulated generation and demand.

`REQ-ELE-005`
The electrical model shall support instructor injection of a station power disturbance.

`REQ-ELE-006`
The electrical model shall display the availability of modeled backup sources.

`REQ-ELE-007`
The electrical model shall allow an operator to request a simulated bus transfer.

`REQ-ELE-008`
The electrical model shall process a transfer request using the latest available source-status data.

`REQ-ELE-009`
The electrical model shall flag a modeled bus as de-energized when its computed supply is absent.

`REQ-ELE-010`
The electrical model shall continue displaying the last measured bus voltage during a communications outage.

`REQ-ELE-011`
The electrical model shall expose source-state changes to the event recorder.

`REQ-ELE-012`
The electrical model shall permit configuration of load priorities for each training scenario.

`REQ-ELE-013`
The electrical model shall represent the delayed start of a simulated backup generator.

`REQ-ELE-014`
The electrical model shall generate a diagnostic message when a requested source is unavailable.

4.9 Reactor Protection System (RPS)

`REQ-RPS-001`
The protection model shall evaluate configured trip predicates at each simulation step.

`REQ-RPS-002`
The protection model shall generate a reactor-trip event when an enabled trip predicate evaluates true.

`REQ-RPS-003`
The protection model shall record the initiating trip predicate and the values used in its evaluation.

`REQ-RPS-004`
The protection model shall provide an operator-visible indication of the current trip state.

`REQ-RPS-005`
The protection model shall accept simulated manual trip commands from an authorized operator session.

`REQ-RPS-006`
The protection model shall allow instructor scripts to disable an individual trip predicate for a training scenario.

`REQ-RPS-007`
The protection model shall evaluate a trip predicate using the last received sensor value when a new sample is missing.

`REQ-RPS-008`
The protection model shall provide a distinct indication for inhibited and enabled trip predicates.

`REQ-RPS-009`
The protection model shall preserve a trip event in the event log after the initiating condition clears.

`REQ-RPS-010`
The protection model shall permit an operator to reset a trip state when the reset command is issued.

`REQ-RPS-011`
The protection model shall publish trip state changes to the control rod and turbine models.

`REQ-RPS-012`
The protection model shall suppress repeated identical trip messages within a configurable display interval.

`REQ-RPS-013`
The protection model shall include the active protection configuration in the scenario export.

`REQ-RPS-014`
The protection model shall allow a scenario author to configure the order in which trip predicates are evaluated.

`REQ-RPS-015`
The protection model shall report a diagnostic event when a required protection input is invalid.

`REQ-RPS-016`
The protection model shall retain the current operating mode when a protection-input diagnostic event occurs.

4.10 Emergency Cooling and Mitigation (ECC)

`REQ-ECC-001`
The emergency cooling model shall represent standby, available, active, and unavailable states.

`REQ-ECC-002`
The emergency cooling model shall compute a simplified coolant-addition estimate when activated.

`REQ-ECC-003`
The emergency cooling model shall display the current emergency cooling state on the operator dashboard.

`REQ-ECC-004`
The emergency cooling model shall accept activation commands from the protection model.

`REQ-ECC-005`
The emergency cooling model shall support a manual activation command from the operator interface.

`REQ-ECC-006`
The emergency cooling model shall allow the operator to cancel an activation request before the next simulation step.

`REQ-ECC-007`
The emergency cooling model shall record activation and deactivation events with timestamps.

`REQ-ECC-008`
The emergency cooling model shall indicate when a modeled supply source is unavailable.

`REQ-ECC-009`
The emergency cooling model shall use the most recently computed coolant-inventory estimate during upstream communication delays.

`REQ-ECC-010`
The emergency cooling model shall support instructor injection of a failed-pump condition.

`REQ-ECC-011`
The emergency cooling model shall publish its simulated flow contribution to the coolant model.

`REQ-ECC-012`
The emergency cooling model shall permit deactivation while a low-inventory advisory remains active.

`REQ-ECC-013`
The emergency cooling model shall provide a post-run timeline of emergency cooling commands.

`REQ-ECC-014`
The emergency cooling model shall label all simulated flow values with engineering units.

4.11 Containment and Environmental Monitoring (CNT)

`REQ-CNT-001`
The containment model shall estimate simulated enclosure pressure and temperature.

`REQ-CNT-002`
The containment model shall represent the effect of configurable leak disturbances.

`REQ-CNT-003`
The containment model shall provide trends for pressure, temperature, and modeled airborne activity.

`REQ-CNT-004`
The containment model shall generate an event when a monitored variable crosses a scenario-defined threshold.

`REQ-CNT-005`
The containment model shall display sensor-quality status alongside monitored values.

`REQ-CNT-006`
The containment model shall retain the most recent environmental measurement when its data source disconnects.

`REQ-CNT-007`
The containment model shall allow instructor-defined environmental disturbances during an exercise.

`REQ-CNT-008`
The containment model shall export a summary of peak values observed in a run.

`REQ-CNT-009`
The containment model shall support configurable sampling intervals for environmental trends.

`REQ-CNT-010`
The containment model shall allow an operator to silence environmental advisory sounds for the current session.

`REQ-CNT-011`
The containment model shall preserve threshold-crossing events in the audit history.

`REQ-CNT-012`
The containment model shall display the current scenario identifier on the monitoring screen.

4.12 Instrumentation and Sensor Data (INS)

`REQ-INS-001`
The instrumentation service shall assign a unique tag to each simulated measurement channel.

`REQ-INS-002`
The instrumentation service shall store each measurement with a value, engineering unit, timestamp, and quality flag.

`REQ-INS-003`
The instrumentation service shall distinguish valid, stale, missing, and invalid measurements.

`REQ-INS-004`
The instrumentation service shall support configurable measurement noise for training scenarios.

`REQ-INS-005`
The instrumentation service shall make current measurement values available through a read-only API.

`REQ-INS-006`
The instrumentation service shall allow a scenario author to inject a fixed sensor bias.

`REQ-INS-007`
The instrumentation service shall publish the last valid measurement when a channel enters a stale state.

`REQ-INS-008`
The instrumentation service shall show measurement-quality indicators in detailed sensor views.

`REQ-INS-009`
The instrumentation service shall log changes in measurement-quality status.

`REQ-INS-010`
The instrumentation service shall permit a display widget to hide quality indicators when compact mode is selected.

`REQ-INS-011`
The instrumentation service shall use a consistent timestamp format across measurement channels.

`REQ-INS-012`
The instrumentation service shall allow an instructor to replace a live simulated measurement with a scripted value.

`REQ-INS-013`
The instrumentation service shall validate that sensor tags are unique within a scenario.

`REQ-INS-014`
The instrumentation service shall retain sensor calibration metadata with the scenario definition.

`REQ-INS-015`
The instrumentation service shall provide the originating model name for each measurement.

`REQ-INS-016`
The instrumentation service shall batch measurements for efficient display updates.

4.13 Alarms, Events, and Notifications (ALM)

`REQ-ALM-001`
The alarm service shall assign a unique identifier to every alarm definition.

`REQ-ALM-002`
The alarm service shall display active alarms with timestamps and originating subsystems.

`REQ-ALM-003`
The alarm service shall support operator acknowledgement of individual alarms.

`REQ-ALM-004`
The alarm service shall preserve acknowledgement history in the run log.

`REQ-ALM-005`
The alarm service shall distinguish active, acknowledged, and cleared alarm states.

`REQ-ALM-006`
The alarm service shall provide filters by subsystem and time interval.

`REQ-ALM-007`
The alarm service shall allow an operator to silence audible notifications for the remainder of a session.

`REQ-ALM-008`
The alarm service shall suppress repeated notifications from the same source during a configurable quiet period.

`REQ-ALM-009`
The alarm service shall display the most recent 100 events without requiring pagination.

`REQ-ALM-010`
The alarm service shall continue collecting events when the operator dashboard is disconnected.

`REQ-ALM-011`
The alarm service shall permit a scenario to disable selected alarm definitions before a run.

`REQ-ALM-012`
The alarm service shall record the actor responsible for changing alarm configuration.

`REQ-ALM-013`
The alarm service shall display a count of unacknowledged active alarms.

`REQ-ALM-014`
The alarm service shall allow an operator to clear an alarm after acknowledging it.

`REQ-ALM-015`
The alarm service shall export the full event history in a machine-readable format.

`REQ-ALM-016`
The alarm service shall preserve the original event timestamp when events arrive late.

4.14 Human-Machine Interface and Operator Commands (HMI)

`REQ-HMI-001`
The operator interface shall present a plant overview with current simulated power, pressure, temperature, and major equipment states.

`REQ-HMI-002`
The operator interface shall display the selected scenario name and elapsed simulation time.

`REQ-HMI-003`
The operator interface shall provide navigation to subsystem-specific screens.

REQ-HMI-004
The operator interface shall indicate whether displayed measurements are live, stale, or unavailable.

REQ-HMI-005
The operator interface shall provide keyboard-accessible navigation for major controls.

REQ-HMI-006
The operator interface shall require confirmation before deleting a saved scenario.

REQ-HMI-007
The operator interface shall permit execution of equipment commands with a single click when quick-command mode is enabled.

REQ-HMI-008
The operator interface shall display a command's acceptance or rejection result.

REQ-HMI-009
The operator interface shall prevent duplicate submission caused by repeated clicks while a command request is pending.

REQ-HMI-010
The operator interface shall allow operators to rearrange noncritical dashboard widgets.

REQ-HMI-011
The operator interface shall provide a searchable list of equipment tags.

REQ-HMI-012
The operator interface shall allow a user to hide warning banners during presentation mode.

REQ-HMI-013
The operator interface shall record commands submitted through the user interface.

REQ-HMI-014
The operator interface shall show units beside numerical process values.

REQ-HMI-015
The operator interface shall display the currently signed-in user.

REQ-HMI-016
The operator interface shall restore the user's preferred dashboard layout on the next login.

4.15 Automatic Control and Setpoint Management (AUT)

REQ-AUT-001
The automatic control service shall support configurable setpoints for selected simulated process variables.

REQ-AUT-002
The automatic control service shall distinguish automatic, manual, and disabled controller modes.

REQ-AUT-003
The automatic control service shall log each setpoint change with the initiating user or scenario script.

REQ-AUT-004
The automatic control service shall enforce configured minimum and maximum setpoint values.

REQ-AUT-005
The automatic control service shall display the active controller mode next to each controlled variable.

`REQ-AUT-006`
The automatic control service shall support a scripted sequence of setpoint changes.

`REQ-AUT-007`
The automatic control service shall use the last valid process value when a fresh measurement is unavailable.

`REQ-AUT-008`
The automatic control service shall continue issuing the previous output command during a brief input-data interruption.

`REQ-AUT-009`
The automatic control service shall publish controller outputs to their destination subsystem models.

`REQ-AUT-010`
The automatic control service shall allow operators to switch a controller from automatic to manual mode.

`REQ-AUT-011`
The automatic control service shall permit setpoint updates while a controller is disabled.

`REQ-AUT-012`
The automatic control service shall include controller parameters in the scenario configuration export.

`REQ-AUT-013`
The automatic control service shall record saturation events when an output reaches a configured limit.

`REQ-AUT-014`
The automatic control service shall restore the most recently saved setpoint after a simulator restart.

4.16 Scenario Authoring and Fault Injection (SCN)

`REQ-SCN-001`
The scenario service shall allow instructors to create, duplicate, edit, and archive simulation scenarios.

`REQ-SCN-002`
The scenario service shall store a human-readable description for each scenario.

`REQ-SCN-003`
The scenario service shall support scheduled disturbances at specified simulation times.

`REQ-SCN-004`
The scenario service shall support event-triggered disturbances based on modeled state changes.

`REQ-SCN-005`
The scenario service shall record the author and last modification time of each scenario.

`REQ-SCN-006`
The scenario service shall validate that referenced subsystem tags exist before starting a run.

`REQ-SCN-007`
The scenario service shall allow multiple disturbances to be scheduled for the same simulation time.

`REQ-SCN-008`
The scenario service shall permit scenario scripts to override normal operating constraints when an instructor option is enabled.

REQ-SCN-009
The scenario service shall show a preview of scheduled disturbances to the instructor.

REQ-SCN-010
The scenario service shall hide instructor-only disturbance details from the trainee interface.

REQ-SCN-011
The scenario service shall allow an instructor to pause a scenario before injecting a new disturbance.

REQ-SCN-012
The scenario service shall store scenario versions with unique identifiers.

REQ-SCN-013
The scenario service shall allow a scenario author to specify expected observations for later assessment.

REQ-SCN-014
The scenario service shall export scenario definitions in a documented structured format.

REQ-SCN-015
The scenario service shall reject scenarios that reference unknown model identifiers.

REQ-SCN-016
The scenario service shall support deterministic replay when identical seeds and initial conditions are supplied.

4.17 Data Storage, Logging, and Replay (DAT)

REQ-DAT-001
The data service shall store run metadata, scenario version, user identifier, and start and end timestamps.

REQ-DAT-002
The data service shall record time-series samples for configured process variables.

REQ-DAT-003
The data service shall retain operator commands and system events in chronological order.

REQ-DAT-004
The data service shall export run data to CSV and JSON formats.

REQ-DAT-005
The data service shall permit users to search prior runs by scenario and date.

REQ-DAT-006
The data service shall associate each stored sample with its originating subsystem.

REQ-DAT-007
The data service shall buffer samples temporarily when persistent storage is unavailable.

REQ-DAT-008
The data service shall discard buffered noncritical trend samples after a configurable retention limit is reached.

REQ-DAT-009
The data service shall preserve command and trip events when reducing stored trend resolution.

REQ-DAT-010
The data service shall support replay of a saved run without modifying the original run record.

REQ-DAT-011
The data service shall label replayed data as historical rather than live.

REQ-DAT-012
The data service shall permit an administrator to delete runs that have passed the configured retention period.

REQ-DAT-013
The data service shall include model versions and configuration hashes in run exports.

REQ-DAT-014
The data service shall record the cause of a run termination when available.

REQ-DAT-015
The data service shall allow users to annotate events during post-run review.

REQ-DAT-016
The data service shall present timestamps in a consistent time reference.

4.18 User Accounts and Access Control (SEC)

REQ-SEC-001
The account service shall authenticate users before permitting access to saved scenarios and run histories.

REQ-SEC-002
The account service shall support trainee, instructor, and administrator roles.

REQ-SEC-003
The account service shall store passwords using a one-way password-hashing algorithm.

REQ-SEC-004
The account service shall record successful and failed login attempts.

REQ-SEC-005
The account service shall allow users to change their own passwords.

REQ-SEC-006
The account service shall require instructor permissions to modify scenario disturbance schedules.

REQ-SEC-007
The account service shall allow a shared demonstration account to access the operator dashboard.

REQ-SEC-008
The account service shall terminate inactive sessions after a configurable interval.

REQ-SEC-009
The account service shall provide administrators with a list of active user sessions.

REQ-SEC-010
The account service shall permit instructors to start training runs without an additional approval workflow.

REQ-SEC-011
The account service shall restrict account-management operations to administrators.

REQ-SEC-012
The account service shall allow users to export data from runs they are permitted to view.

REQ-SEC-013
The account service shall record role changes with the initiating administrator.

REQ-SEC-014
The account service shall support revocation of active sessions after an account is disabled.

4.19 Software Interfaces and Communications (NET)

REQ-NET-001
The interface service shall publish subsystem state updates using a documented message schema.

REQ-NET-002
The interface service shall assign sequence numbers to state-update messages.

REQ-NET-003
The interface service shall detect missing or out-of-order messages.

REQ-NET-004
The interface service shall provide a health endpoint for each simulated subsystem service.

REQ-NET-005
The interface service shall retry failed noncritical telemetry deliveries within configured limits.

REQ-NET-006
The interface service shall permit command delivery using the most recent known destination status.

REQ-NET-007
The interface service shall reject messages with malformed required fields.

REQ-NET-008
The interface service shall record interface errors with timestamps and source identifiers.

REQ-NET-009
The interface service shall maintain a configurable timeout for inter-service requests.

REQ-NET-010
The interface service shall allow the operator interface to reconnect after temporary network loss.

REQ-NET-011
The interface service shall provide an explicit stale-data flag when a subsystem stops publishing.

REQ-NET-012
The interface service shall support independent versioning of message schemas.

REQ-NET-013
The interface service shall permit asynchronous delivery of noncritical display updates.

REQ-NET-014
The interface service shall document the behavior of each command endpoint when a request is repeated.

4.20 Verification, Validation, and Experiment Support (VNV)

REQ-VNV-001
The verification framework shall provide automated tests for documented subsystem interfaces.

REQ-VNV-002
The verification framework shall compare selected simulation outputs against scenario-specific reference traces.

REQ-VNV-003
The verification framework shall record the model version and configuration used for each test.

REQ-VNV-004
The verification framework shall provide pass, fail, and inconclusive outcomes for automated checks.

REQ-VNV-005
The verification framework shall allow instructors to define expected trainee observations.

REQ-VNV-006
The verification framework shall export test results in a machine-readable format.

REQ-VNV-007
The verification framework shall verify that all requirement identifiers in this specification are unique.

REQ-VNV-008
The verification framework shall support repeated execution of a scenario using a fixed random seed.

REQ-VNV-009
The verification framework shall measure execution time for each simulation step.

REQ-VNV-010
The verification framework shall report when an expected event is absent from a run.

REQ-VNV-011
The verification framework shall permit a test operator to mark a failed test as accepted with a written justification.

REQ-VNV-012
The verification framework shall retain evidence of overridden test outcomes.

REQ-VNV-013
The verification framework shall provide a traceability field linking a test case to one or more requirement identifiers.

REQ-VNV-014
The verification framework shall include boundary-value tests for configurable input ranges.

REQ-VNV-015
The verification framework shall detect differences between replayed and original event sequences.

REQ-VNV-016
The verification framework shall produce a run-level summary for comparative experiments.

4.21 Performance, Reliability, and Recovery (PER)

REQ-PER-001
The simulator shall complete each nominal simulation step within a configurable execution-time budget on the reference workstation.

REQ-PER-002
The simulator shall display an indication when the execution-time budget is exceeded.

REQ-PER-003
The simulator shall recover the operator interface after a browser refresh without deleting the current run.

REQ-PER-004
The simulator shall support checkpoint creation at configured simulation-time intervals.

REQ-PER-005
The simulator shall allow recovery from the most recently saved checkpoint.

REQ-PER-006
The simulator shall record unexpected subsystem restarts in the event log.

REQ-PER-007
The simulator shall continue running unaffected subsystem models when a noncritical reporting service fails.

REQ-PER-008
The simulator shall restore the last saved operator preferences after an application restart.

REQ-PER-009
The simulator shall permit checkpoint restoration even when the saved checkpoint contains unresolved advisory alarms.

REQ-PER-010
The simulator shall reject corrupted checkpoint files before restoring state.

REQ-PER-011
The simulator shall report unavailable dependent services in the diagnostic dashboard.

REQ-PER-012
The simulator shall provide a configurable upper bound on stored trend samples per run.

REQ-PER-013
The simulator shall allow an instructor to terminate an unresponsive run.

REQ-PER-014
The simulator shall preserve the run termination reason in the audit record.

4.22 Documentation, Training, and Usability (DOC)

REQ-DOC-001
The system shall provide an operator guide describing each simulated subsystem and its controls.

REQ-DOC-002
The system shall provide a glossary of plant-model variables and engineering units.

REQ-DOC-003
The system shall include an instructor guide for scenario creation and disturbance injection.

REQ-DOC-004
The system shall provide a help link from each major operator screen.

REQ-DOC-005
The system shall document known model simplifications and assumptions.

REQ-DOC-006
The system shall label the application as a research and training simulation.

REQ-DOC-007
The system shall display the active software version in an About screen.

`REQ-DOC-008`
The system shall provide a keyboard-accessible method to acknowledge alarms.

`REQ-DOC-009`
The system shall present error messages in language understandable to a trained user.

`REQ-DOC-010`
The system shall allow an instructor to attach learning objectives to a scenario.

`REQ-DOC-011`
The system shall provide a printable post-run report.

`REQ-DOC-012`
The system shall retain a revision history for this requirements specification.

5 External Interfaces

5.1 Operator Interface

The user interface presents plant-state displays, control actions, alarms, scenario context, and run status. Detailed screen behavior is governed by the HMI and ALM requirements.

5.2 Model and Service Interfaces

Subsystems exchange time-stamped state messages through documented interfaces. Command and telemetry schemas are versioned separately. The INS and NET requirements govern data quality and delivery behavior.

5.3 Persistent Data Interfaces

The storage layer maintains scenario definitions, user metadata, run traces, event histories, and test results. Data retention and export are governed by DAT and SEC requirements.

6 Verification and Acceptance Approach

6.1 Requirement-Level Evidence

For each requirement, the project shall maintain a separate verification record containing its identifier, proposed test or inspection method, test environment, expected observation, actual observation, outcome, and evidence reference. This section defines the documentation approach and does not change the individual requirements above.

6.2 Suggested Verification Methods

- **Inspection:** confirm that screens, documents, schemas, and configurations contain the required fields and controls.
- **Functional testing:** exercise user commands, mode changes, scenario transitions, and event recording.
- **Simulation testing:** apply defined input conditions and compare state transitions or time-series outputs with scenario reference expectations.

- **Fault-injection testing:** simulate missing data, stale values, unavailable services, or injected component failures.
- **Replay and audit testing:** verify trace consistency, timestamps, user attribution, and reproducibility.
- **Usability testing:** assess navigation, readability, keyboard access, and clarity of feedback.

6.3 Traceability

A requirements-to-tests traceability matrix should be maintained outside this source specification. Any subsequent engineering assessment, classification, or corrective recommendation should be recorded separately to avoid altering the source requirements used in comparative studies.

7 Configuration and Change Control

Each document revision shall record its version, date, author, and summary of changes. Requirement identifiers shall not be reused for unrelated behaviors. Changed requirements should retain their identifiers only when their intent remains substantially the same; otherwise, a new identifier should be assigned and the superseded statement recorded in revision history.

8 Glossary

Scenario
A configured initial condition, sequence of disturbances, and exercise metadata.

Simulation step
One discrete update interval of the numerical models.

Trip
A modeled protection event that changes the simulated operating state.

Stale measurement
A measurement whose timestamp exceeds the configured freshness interval.

Instructor
A user who configures exercises and disturbances.

Trainee
A user who operates the simulated plant during an exercise.

Run
One execution of a scenario with recorded inputs and outputs.

9 Document Revision History

Version	Date	Description
1.0	October 2026	Initial research simulation specification